

Santiago Martínez Novoa

Systems and Computing Engineer

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EDUCATION

Universidad de los Andes

M.Sc. Information Engineering

Bogotá, Colombia

2025 – 2026 (*To be conferred*)

Universidad de los Andes

B.Sc. Systems and Computing Engineering (Cum Laude), Minor in French

Bogotá, Colombia

2021 – 2024

EXPERIENCE

Mitacs Globalink Research Intern - Collaborating Researcher

Ontario Tech University

May 2026 – Present

Oshawa, Canada

- Co-authored a Findings of EMNLP 2026 paper on adaptive language sampling; leading its follow-up on when to stop cross-lingual realignment, under review at ARR
- Co-developing M2-ALIGN: extending vision-language models to 11 languages while preserving their multilingual reasoning

Data Scientist

EdTech Node, Universidad de los Andes

Mar 2026 – Present

Bogotá, Colombia

- Lead the unit's adoption of data governance and quality policies, enabling university-wide data products such as academic percentiles and dropout indicators
- Lead the team's research line: dropout-prediction ML models behind 2 peer-reviewed publications (AIED & WEEF 2026)

Graduate Teaching Assistant

Universidad de los Andes

Jun 2025 – Present

Bogotá, Colombia

- Designed the hands-on curriculum for two sequential master's courses on RAG systems and AI agents, reaching 200+ students across 3 semesters
- Built tutorials, projects, and RAGAS-based evaluation pipelines with LangChain, LangGraph, and LangSmith

Data Scientist - Graduate Research Assistant

Vice Dean's Office of Student Affairs, Universidad de los Andes

Jan 2025 – Mar 2026

Bogotá, Colombia

- Implemented complete medallion architecture in Databricks, fully automating institutional data pipelines
- Optimized ETL pipeline performance by 87% by migrating from Pandas to Polars, reducing processing time from 15 to 2 minutes
- Replaced external counseling platform, saving the university ~\$100K USD annually
- Co-implemented system alerting over 1000 critical student cases

Generative Artificial Intelligence for Research - Research Assistant

Universidad de los Andes

Jan 2025 – Jun 2025

Bogotá, Colombia

- Built locally-deployed RAG systems using Ollama and Gemini API for domain-specific research questions
- Designed LLM evaluation architecture using Prometheus, BERT score and OpenAI API for response quality assessment
- Developed custom AI agents to optimize university research processes
- Automated student assessment using LLMs aligned with rubrics and expected answers

Machine Learning Intern

Precia PPV S.A

Aug 2024 – Dec 2024

Bogotá, Colombia

- Developed MLP model reducing financial calculation time by 70% for client applications
- Implemented end-to-end AWS infrastructure with automated ETL pipelines using Glue and Lambda
- Presented our work on models and AWS infrastructure at Analytics Forum 2025

Summer Research Intern

College of Computing and Information Science, Cornell University

Jun 2024 – Aug 2024

Ithaca, United States

- Designed Data Discovery architecture to explore and modify datasets in various formats
- Developed upload and scoring system through semantic search across multiple databases

TECHNICAL SKILLS

- *Programming Languages:* Python, SQL, JavaScript, TypeScript
- *Machine Learning & AI:* Scikit-learn, Keras, PyTorch, TensorFlow
- *LLM & Natural Language Processing:* Transformers, BERT, LangChain, LangGraph, LangSmith, Ollama
- *Data Processing:* Pandas, Polars, Databricks
- *Cloud & DevOps:* AWS, Azure Functions, Azure Blob Storage
- *Databases:* PostgreSQL, MongoDB, ChromaDB, Pinecone